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Overview and Descriptive Statistics

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1.3 Measures of Location

Measures of Location

Visual summaries of data are excellent tools for obtaining preliminary impressions and insights. More formal data analysis often requires the calculation and interpretation of numerical summary measures.

That is, from the data we try to extract several summarizing numbers—numbers that might serve to characterize the data set and convey some of its salient features. Our primary concern will be with **numerical data**; some comments regarding categorical data appear at the end of the section.

Measures of Location

Suppose, then, that our data set is of the form $X_1, X_2, \dots, X_n$, where each X_i is a number.

What features of such a set of numbers are of most interest and deserve emphasis?

One important characteristic of a set of numbers is its location, and in particular its center.

This section presents methods for describing the location of a data set.



The Mean

The Mean

For a given set of numbers $x_1, x_2, \dots, x_n$, the most familiar and useful measure of the center is the *mean*, or *arithmetic average* of the set.

Because we will almost always think of the x_i 's as *constituting a sample*, we will often refer to the arithmetic average as the *sample mean* and denote it by $\bar{x}$.

The Mean

The **sample mean** $\bar{x}$ of observations $x_1, x_2, \dots, x_n$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

The numerator of $\bar{x}$ can be written more informally as Σx_i , where the summation is over all sample observations.

The Mean

The numerator of $\bar{x}$ can be written more informally as $\sum x_i$, where the summation is over all sample observations.

For reporting $\bar{x}$, we **recommend using decimal accuracy of one digit more than the accuracy** of the x_i 's.

Thus if observations are stopping distances with ,

$$x_1 = 125,$$

$$x_2 = 131, \text{ and so on,}$$

we might have $\bar{x} = 127.3$ ft.

Example

Caustic stress corrosion cracking of iron and steel has been studied because of failures around rivets in steel boilers and failures of steam rotors.

Consider the accompanying **observations on x = crack length (μm)** as a result of constant load stress corrosion tests on smooth bar tensile specimens for a fixed length of time. (The data is consistent with a histogram and summary quantities from the article “**On the Role of Phosphorus in the Caustic Stress Corrosion Cracking of Low Alloy Steels,**” *Corrosion Science*, 1989: 53–68.)

Example

cont'd

$$x_1 = 16.1 \quad x_2 = 9.6 \quad x_3 = 24.9 \quad x_4 = 20.4 \quad x_5 = 12.7$$

$$x_6 = 21.2 \quad x_7 = 30.2 \quad x_8 = 25.8 \quad x_9 = 18.5 \quad x_{10} = 10.3$$

$$x_{11} = 25.3 \quad x_{12} = 14.0 \quad x_{13} = 27.1 \quad x_{14} = 45.0 \quad x_{15} = 23.3$$

$$x_{16} = 24.2 \quad x_{17} = 14.6 \quad x_{18} = 8.9 \quad x_{19} = 32.4 \quad x_{20} = 11.8$$

$$x_{21} = 28.5$$

Example

cont'd

Figure 1.14 shows a stem-and-leaf display of the data; a crack length in the low 20s appears to be “typical.”

0H	96	89				
1L	27	03	40	46	18	
1H	61	85				
2L	49	04	12	33	42	
2H	58	53	71	85		
3L	02	24				
3H						
4L						
4H	50					

Stem: tens digit
Leaf: one and tenths digit

A stem-and-leaf display of the crack-length data

Example

cont'd

With $\sum x_i = 444.8$ the **sample mean** is

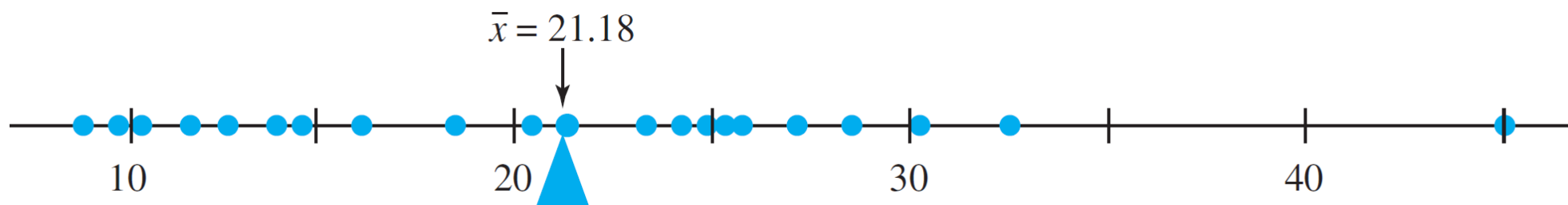
$$\begin{aligned}\bar{x} &= \frac{444.8}{21} \\ &= 21.18\end{aligned}$$

a value consistent with information conveyed by the stem-and-leaf display.

The Mean

A physical interpretation of $\bar{x}$ demonstrates **how it measures the location (center) of a sample**. Think of drawing and scaling a horizontal measurement axis, and then represent each sample observation by a 1-lb weight placed at the corresponding point on the axis.

The only point at which a fulcrum can be placed to balance the system of weights is the point corresponding to the value of $\bar{x}$.



The mean as the balance point for a system of weights

The Mean

Just as $\bar{x}$ represents the average value of the observations in a sample, the average of all values in the population can be calculated. This average is called the **population mean** and is denoted by the Greek letter μ .

When there are N values in the population (a finite population), then

$$\mu = (\text{sum of the } N \text{ population values})/N$$

Just as $\bar{x}$ is an interesting and important measure of sample location, μ is an interesting and important (often the most important) characteristic of a population.

The Mean

The mean suffers from one deficiency: Its value can be greatly affected by the presence of even a single outlier (unusually large or small observation).

In the previous Example, the value $x_{14} = 45.0$ is obviously an outlier.

$\bar{x} = 399.8/20 = 19.99$; without the outlier

$\bar{x} = 399.8/20 = 21.18$; with the outlier

the outlier increases the mean by more than 1 μm .

If the 45.0 μm were replaced by 295.0 μm , then:

$\bar{x} = 694.8/21 = 33.09$,

which is larger than all but one of the observations!



The Median

The Median

The word *median* is synonymous with “middle,” and the sample median is indeed the middle value once the observations are ordered from smallest to largest.

When the observations are denoted by $x_1, \dots, x_n$, we will use the symbol $\tilde{x}$ to represent the sample median.

The symbol $\tilde{x}$ denotes the **sample median**.

The Median

Definition: The **sample median** is obtained by first **ordering** the n observations from smallest to largest (with any **repeated values included** so that every sample observation appears in the ordered list). Then,

The Median

cont'd

The **sample median**:

The **sample median** is obtained by first ordering the n observations from smallest to largest (with any repeated values included so that every sample observation appears in the ordered list). Then,

$$\tilde{x} = \begin{cases} \text{The single middle value if } n \text{ is odd} & = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ ordered value} \\ \text{The average of the two middle values if } n \text{ is even} & = \text{average of } \left(\frac{n}{2} \right)^{\text{th}} \text{ and } \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ ordered values} \end{cases}$$

Example

People not familiar with classical music might tend to believe that a composer's instructions for playing a particular piece are so specific that the duration would not depend at all on the performer(s).

However, there is typically plenty of room for interpretation, and orchestral conductors and musicians take full advantage of this.

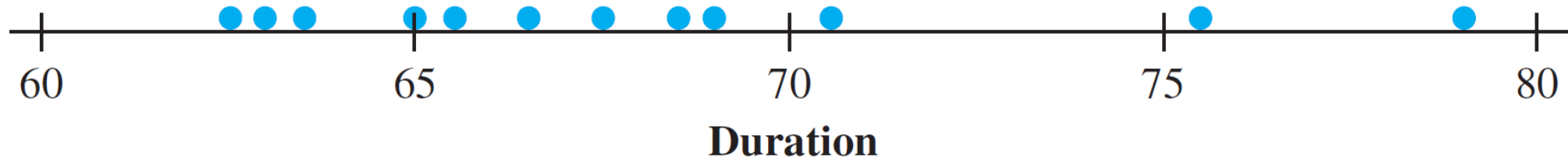
Example

cont'd

The author selected **a sample of 12 recordings** of Beethoven's Symphony #9 (the "Choral," a stunningly beautiful work), yielding the following durations (min) listed in increasing order:

62.3 62.8 63.6 65.2 65.7 66.4 67.4 68.4 68.8 70.8
75.7 79.0

Here is a dotplot of the data:



Dotplot of the data from Example 14

Example

cont'd

Since $n = 12$ is even, the sample median is the average of the $n/2 = 6^{\text{th}}$ and $(n/2 + 1) = 7^{\text{th}}$ values from the ordered list:

$$\tilde{x} = \frac{66.4 + 67.4}{2} = 66.90$$

Example for odd n :

Eliminate 79.0 from the sample,

the resulting sample median for the $n = 11$ observations would have been the single middle value 66.4 (the $[n + 1]/2 = 6^{\text{th}}$ ordered value, i.e. the 6^{th} value in from either end of the ordered list).

Example

cont'd

The sample mean:

$$\bar{x} = \Sigma x_i / n = 816.1 / 12 = 68.01,$$

$$\tilde{x} = \frac{66.4 + 67.4}{2} = 66.90$$

The sample mean is larger than the median.

The mean is pulled out a bit relative to the median because the sample “stretches out” somewhat more on the upper end than on the lower end.

The Median

The data in the previous Example illustrates an important property of $\tilde{x}$:

(in contrast to $\bar{x}$) **The sample median is very insensitive to outliers.**

If, for example, we increased the two largest x_i s from 75.7 and 79.0 to 85.7 and 89.0, respectively, $\tilde{x}$ would be unaffected.

Thus, in the treatment of outlying data values, $\bar{x}$ and $\tilde{x}$ are at opposite ends of a spectrum. **Both quantities describe where the data is centered, but they will not in general be equal** because they focus on different aspects of the sample.

The Median

Analogous to $\tilde{x}$ as the middle value in the sample is a middle value in the population, the **population median**, denoted by $\tilde{\mu}$. As with $\bar{x}$ and μ , we can think of using the sample median $\tilde{x}$ to make an inference about $\tilde{\mu}$.

Sample mean

$\bar{x}$

Population mean

μ

Sample median

$\tilde{x}$

Population median

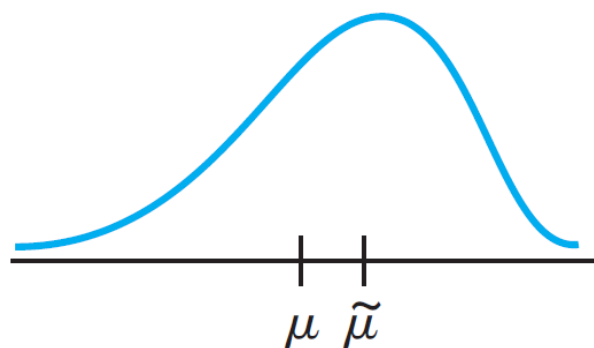
$\tilde{\mu}$.

In the previous Example, we might use $\tilde{x} = 66.90$ as an estimate of the median time for the population of all recordings. A median is often used to describe income or salary data (because it is not greatly influenced by a few large salaries).

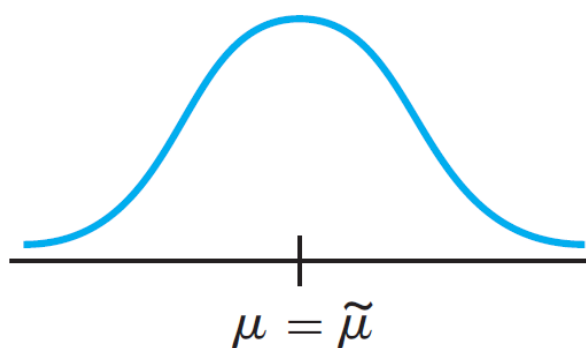
The Median

If the median salary for a sample of engineers were $\tilde{x} = \$66,416$, we might use this as a basis for concluding that the median salary for all engineers exceeds \$60,000.

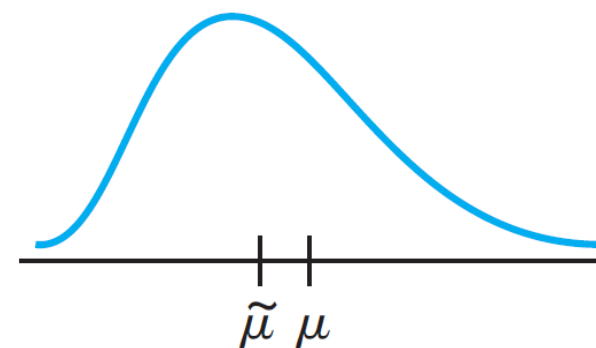
The population mean μ and $\tilde{\mu}$. median will not generally be identical. If the population distribution is positively or negatively skewed, as pictured below, then $\mu \neq \tilde{\mu}$.



(a) Negative skew



(b) Symmetric



(c) Positive skew

Three different shapes for a population distribution



Other Measures of Location: Quartiles, Percentiles, and Trimmed Means

Quartiles, Percentiles, and Trimmed Means

The median (population or sample) divides the data set into two parts of equal size. To obtain finer measures of location, we could divide the data into more than two such parts.

Quartiles divide the data set into four equal parts, with the observations above the third quartile constituting the upper quarter of the data set, **the second quartile being identical to the median**, and the first quartile separating the lower quarter from the upper three-quarters.

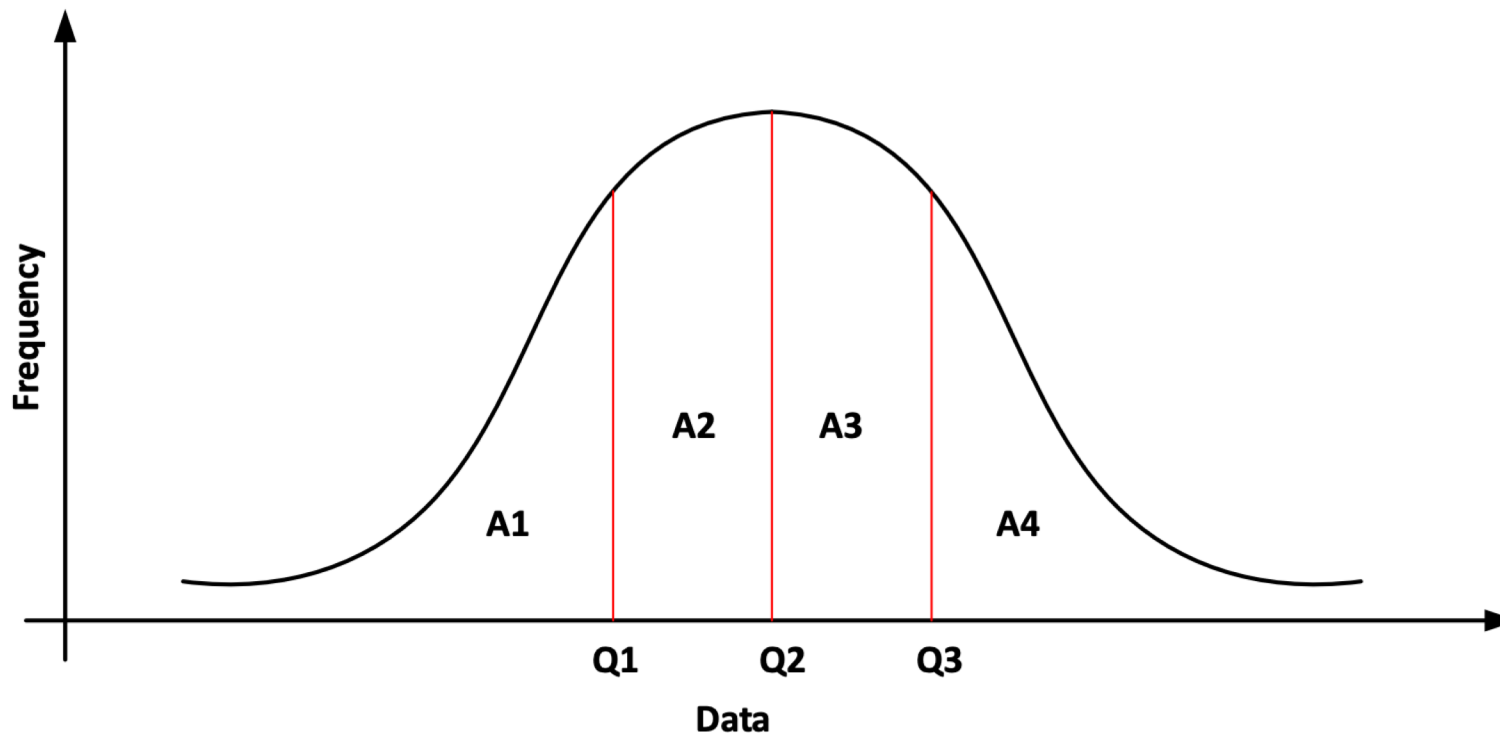
Quartiles, Percentiles, and Trimmed Means

Similarly, a data set (sample or population) can be even more finely divided using **percentiles**; the 99th percentile separates the highest 1% from the bottom 99%, and so on.

Unless the number of observations is a multiple of 100, care must be exercised in obtaining percentiles.

Quartiles, Percentiles, and Trimmed Means

Quartiles:



Q1 = first quartile
Q2 = second quartile
Q3 = third quartile
Where:
 $A1 = A2 = A3 = A4$

Quartiles, Percentiles, and Trimmed Means

The mean is quite sensitive to a single outlier, whereas the median is impervious to many outliers. Since extreme behavior of either type might be undesirable, we briefly consider **alternative measures** that are neither as sensitive as $\bar{x}$ nor as insensitive as $\tilde{x}$.

To motivate these alternatives, note that $\bar{x}$ and $\tilde{x}$ are at opposite extremes of the same “family” of measures.

The **mean** is the **average of all the data**, whereas the **median** results from **eliminating all but the middle one** or two values and then averaging.

Quartiles, Percentiles, and Trimmed Means

To paraphrase, **the mean involves trimming 0%** from each end of the sample, whereas for **the median the maximum possible amount is trimmed from each end.**

A **trimmed mean** is a compromise between $\bar{x}$ and $\tilde{x}$. A 10% trimmed mean, for example, would be computed by eliminating the smallest 10% and the largest 10% of the sample and then averaging what remains.

Example

The production of Bidri is a traditional craft of India. Bidri wares (bowls, vessels, and so on) are cast from an alloy containing primarily zinc along with some copper.

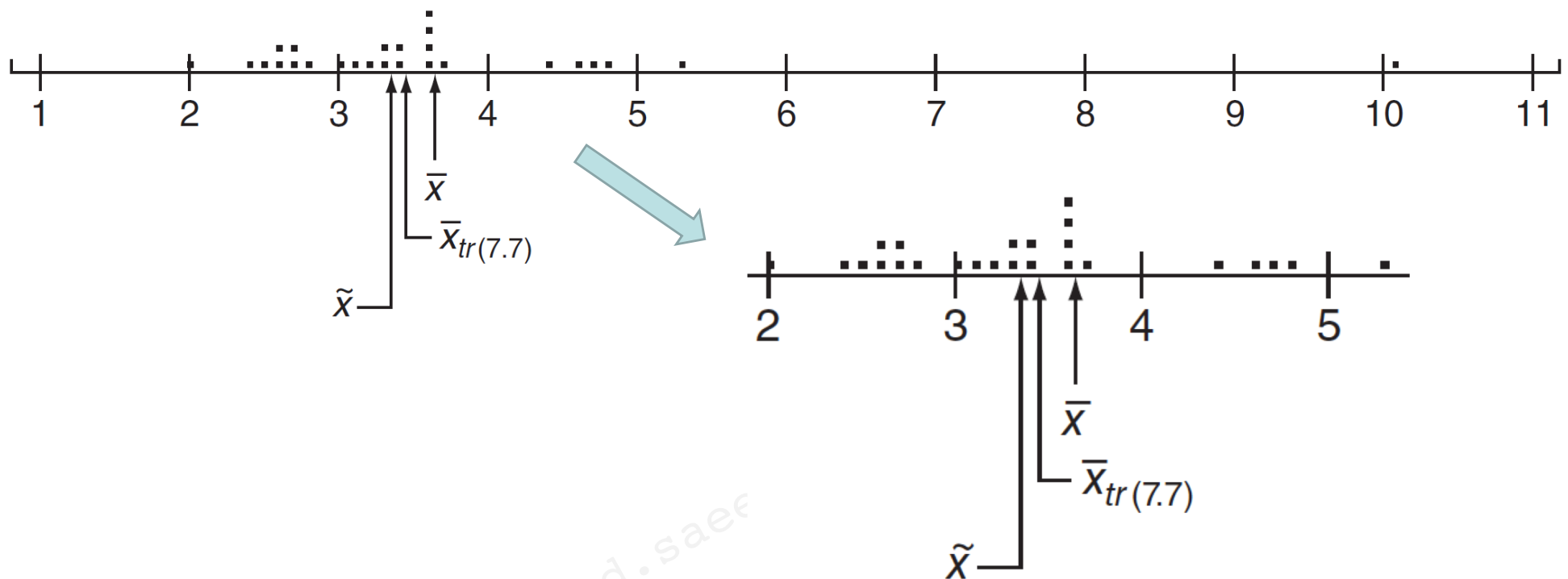
Consider the following observations on copper content (%) for a sample of Bidri artifacts in London's Victoria and Albert Museum ([“Enigmas of Bidri,” *Surface Engr.*, 2005: 333–339](#)), listed in increasing order:

2.0	2.4	2.5	2.6	2.6	2.7	2.7	2.8	3.0	3.1	3.2	3.3	3.3
3.4	3.4	3.6	3.6	3.6	3.6	3.7	4.4	4.6	4.7	4.8	5.3	10.1

Example

cont'd

A dotplot of the data is displayed. A prominent feature is the single outlier at the upper end; the distribution is somewhat sparser in the region of larger values than is the case for smaller values.



Dotplot of copper contents

Example

cont'd

The sample mean and median are 3.65 and 3.35, respectively.

A trimmed mean with a trimming percentage of $100(2/26) = 7.7\%$ results from eliminating the two smallest and two largest observations; this gives

$$\bar{x}_{tr(7.7)} = 3.42.$$

Trimming here eliminates the larger outlier and so pulls the trimmed mean toward the median.



Categorical Data and Sample Proportions

[grad.saeed@gmail.com](https://github.com/saeedgrad)

Categorical Data and Sample Proportions

When the data is **categorical**, a **frequency distribution or relative frequency distribution provides an effective tabular summary of the data**. The natural numerical summary quantities in this situation are the individual frequencies and the relative frequencies.

For example, if a survey of individuals who own digital cameras is undertaken to study brand preference, then each individual in the sample would identify the brand of camera that he or she owned, from which we could count the number owning Canon, Sony, Kodak, and so on.

Categorical Data and Sample Proportions

Consider sampling a dichotomous population—one that consists of only **two categories** (such as voted or did not vote in the last election, does or does not own a digital camera, etc.).

If we let

x denote the number in the sample falling in category 1, the number in category 2 is $n - x$.

Relative frequency or *sample proportion* in category 1: x/n .

Sample proportion in category 2:

$1 - x/n$.

Categorical Data and Sample Proportions

Let's denote a response that falls in category 1 by a 1 and a response that falls in category 2 by a 0. A sample size of $n = 10$ might then yield the responses 1, 1, 0, 1, 1, 1, 0, 0, 1, 1. The sample mean for this numerical sample is (since number of 1s = $x = 7$)

$$\frac{x_1 + \cdots + x_n}{n} = \frac{1 + 1 + 0 + \cdots + 1 + 1}{10} = \frac{7}{10} = \frac{x}{n} = \text{sample proportion}$$

More generally, *focus attention on a particular category and code the sample results so that a 1 is recorded for an observation in the category and a 0 for an observation not in the category.*

Categorical Data and Sample Proportions

Then the sample proportion of observations in the category is the sample mean of the sequence of 1s and 0s. Thus a sample mean can be used to summarize the results of a categorical sample.

These remarks also apply to situations in which categories are defined by grouping values in a numerical sample or population (e.g., we might be interested in knowing whether individuals have owned their present automobile for at least 5 years, rather than studying the exact length of ownership).

Categorical Data and Sample Proportions

Analogous to the sample proportion x/n of individuals or objects falling in a particular category, let p represent the proportion of those in the entire population falling in the category.

As with x/n , p is a quantity between 0 and 1, and while x/n is a sample characteristic, p is a characteristic of the population.

Categorical Data and Sample Proportions

The relationship between the two parallels the relationship between $\tilde{x}$ and $\tilde{\mu}$ and between $\bar{x}$ and μ .

In particular, we will subsequently **use x/n (sample characteristic) to make inferences about p (population characteristic)** .

If, for example, **a sample of 100 car owners reveals that 22 owned their car at least 5 years**, then we might use $22/100 = .22$ as a point estimate of **the proportion of all owners** who have owned their car at least 5 years.

With k categories ($k > 2$), we can use the k sample proportions to answer questions about the population proportions $p_1, \dots, p_k$.